# Zakhar Pashkin


## Senior Computer Vision Engineer | AI Product Engineer | OCR, Segmentation, Detection, VLM/LLM Systems

Yerevan, Armenia | kaisenaiko@gmail.com | zack-dev-cm.github.io | linkedin.com/in/zakhar-pashkin-a524a6163 | github.com/zack-dev-cm | t.me/rheuiii


### SUMMARY


Senior Computer Vision Engineer and AI Product Engineer with 7+ years shipping OCR, segmentation, detection, multimodal search, VLM/LLM workflows, backend services, and production ML systems. Strongest where model quality, product behavior, operator workflow, and launch reliability need one owner: dataset quality, model selection, API delivery, UI surfaces, monitoring, review gates, leak checks, and rollback-ready releases.


### CORE STACK

**AI products:** VLM/LLM workflows, OpenAI APIs, agent automation, Telegram mini apps, Chrome extensions, review gates

**Computer vision:** OCR, segmentation, detection, landmarking, tracking, face texture analysis, neural search, multimodal CV

**Modeling:** PyTorch, TensorFlow/Keras, OpenCV, YOLO, MMDetection, CLIP/VLMs, Transformers, ONNX conversion

**Edge/mobile:** ONNX, TFLite, CoreML, TensorRT, PyTorch Mobile, quantization, mobile inference profiling, Flutter/Android/iOS integration

**Production:** FastAPI, Tornado, REST/gRPC, Docker, Kubernetes, GitHub Actions, GCP, AWS, Cloud Run, monitoring, CI/CD, MLOps

## EXPERIENCE

### Senior Computer Vision Engineer & AI Product Engineer

Jun 2022 - Present

Various Startups - selected startup, independent, and direct client engagements

- Delivered AI product systems across automation, segmentation, detection, OCR, landmarking, face search, neural search, multimodal video search, VLM/LLM workflows, data pipelines, and edge deployment.
- Built or modernized models, APIs, operator workflows, and user-facing product surfaces where scale, latency, and release reliability mattered.
- Optimized PyTorch/TensorFlow models for ONNX, TFLite, and CoreML deployment, with mobile inference targets up to 50 to 60 FPS in OCR and segmentation flows.
- Reduced labeling cost by up to 80% with active-learning and auto-annotation pipelines; released tooling around interactive segmentation and OCR workflows.
- Delivered FastAPI/Tornado services, Docker/Kubernetes deployments, GCP/AWS infrastructure, CI/CD, and monitoring for production ML, AI automation, and launch workflows.
- Shipped public OpenClaw/ClawHub release systems, including a downloads tracker showing 3,992 public ClawHub downloads across 11 packages as of 2026-05-15.
- Added review gates for high-risk AI delivery: benchmark checks, human approvals, browser evidence, public-surface leak checks, rollback criteria, and promotion decisions before release.

### Computer Vision Engineer

Jun 2019 - Jun 2022

CFT - Akademgorodok, Novosibirsk

- Developed classification, object detection, OCR, segmentation, and document-recognition algorithms for fintech and mobile products.
- Optimized energy-meter OCR and segmentation models for stable 50 to 60 FPS on mid-range smartphones.
- Built masked face anti-spoofing and Cyrillic offline signature-verification research systems with deployment-focused error analysis.
- Improved specialized mobile-GPU model quality with neural architecture search and deployment-focused tuning.
- Implemented CLIP pre-validation and Toloka automation to reduce labeling cost and review load through click-driven segmentation workflows.

### Computer Vision Project Mentor

Nov 2019 - Jun 2022

CFT apprenticeship corporate program - Part-time

- Mentored pre-internship engineers on production-style CV projects including mobile energy-meter recognition, crowdsourcing with neural networks, and Cyrillic handwriting OCR.
- Reviewed model choices, dataset quality, error analysis, and deployment constraints so student projects could pass engineering review rather than remain demos.

### Machine Learning Project Mentor

Oct 2018 - Oct 2019

School of AI - Remote / San Francisco Bay Area community

- Built and supported ML/DL software for analytics and online education, including Flutter product work and GCP infrastructure.
- Helped secure and operate Google Cloud support for a global AI education platform while coordinating remote contributors and volunteers.

## SELECTED PROJECT EVIDENCE

**Fast OCR ONNX Inference Server**
Three-stage OCR API: line segmentation, word segmentation, CRNN text recognition, Docker packaging, and JSON boxes/text responses.

**Multimodal Video Search Platform**
Keyframes, ASR/OCR, objects, faces, visual embeddings, transcript embeddings, Qdrant/Postgres retrieval, and quality-agent checks.

**Full-Face Wrinkle Segmentation Lab**
Cosmetic face-region masks, YOLO segmentation, skeletonized line traces, overlays, CSV outputs, timing events, and visual QA gates.

**Dermaself Flutter Skin Analysis App**
Flutter/Firebase guided capture app with onboarding, questionnaire intake, offline model runtime, ROI gates, and results screens.

**Nutrition Label OCR Mobile Lab**
SwiftUI food detection and nutrition-label OCR using ML Kit, crop-assisted text recognition, table parsing, and structured results.

**CV Repro Lab Skills**
Reproducible CV experiment workflow with benchmark gates, review dashboards, browser validation, and 948 public ClawHub downloads across two packages.

## RECRUITER EVIDENCE CHECKS

**AI product scope:** Owns model, backend, workflow, UI, and launch reliability rather than stopping at notebooks or demos.

**CV depth:** Can explain data, metrics, failure modes, conversion path, and edge/runtime constraints for shipped OCR, segmentation, and detection work.

**Evidence quality:** Public work links to repos, releases, dated counters, and review gates; non-public CV work is summarized without repo or live-service links.

## EDUCATION & CERTIFICATIONS

Siberian Federal University, 2007-2012. Additional professional training includes TensorFlow, Deep Learning and Computer Vision, SQL, JavaScript, Kubernetes on Google Cloud, and PyTorch scholarship work.